

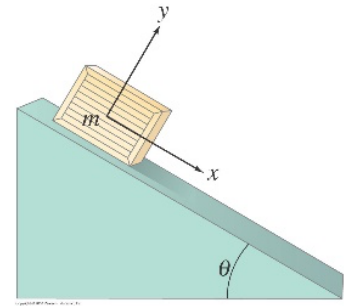
Tutorial 3: Newton's Laws of Motion (2) and Work, Energy, Power

Forces

Q1.

The crate shown in Fig. 5–33 lies on a plane tilted at an angle 25.0° to the horizontal, with coefficient of kinetic friction $\mu_k = 0.19$.

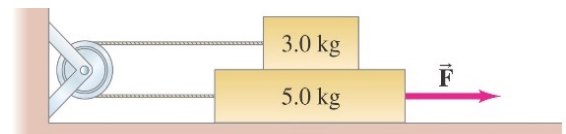
- Determine the acceleration of the crate as it slides down the plane.
- If the crate starts from rest 8.15 m up the plane from its base, what will be the crate's speed when it reaches the bottom of the incline?



Q2.

A 3.0-kg block sits on top of a 5.0-kg block which is on a horizontal surface. The 5.0-kg block is pulled to the right with a force F as shown. The coefficient of static friction between all surfaces is 0.60 and the kinetic coefficient is 0.40.

- What is the minimum value of F needed to move the two blocks?
- If the force is 10% greater than your answer for (a), what is the acceleration of each block?



Q3.

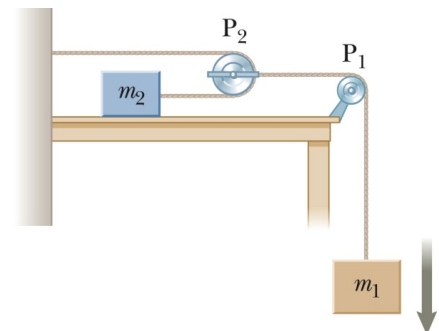
Two crates, of mass 65 kg and 125 kg, are in contact and at rest on a horizontal surface. A 650-N force is exerted on the 65-kg crate. If the coefficient of kinetic friction is 0.18, calculate

- the acceleration of the system, and
- the force that each crate exerts on the other.
- Repeat with the crates reversed.



Q4. An object of mass m_1 hangs from a string that passes over a very light fixed pulley P_1 . The string connects to a second very light pulley P_2 . A second string passes around this pulley with one end attached to a wall and the other to an object of mass m_2 on a frictionless, horizontal table.

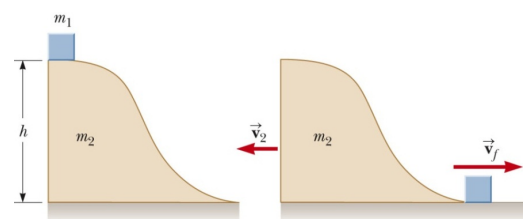
- If a_1 and a_2 are the accelerations of m_1 and m_2 , respectively, what is the relation between these accelerations? Find expressions for
- the tensions in the strings, and
- the accelerations a_1 and a_2 in terms of the masses m_1 and m_2 , and g .



Conservation of Energy, Work-Energy Principle

Q5.

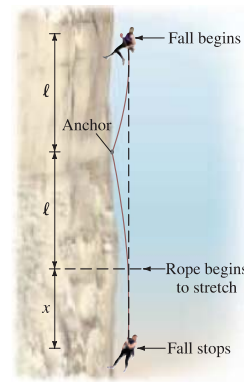
A small block of mass $m_1 = 0.500$ kg is released from rest at the top of a frictionless, curve-shaped wedge of mass $m_2 = 3.00$ kg, which sits on a frictionless, horizontal surface. When the block leaves the wedge, its velocity is measured to be 4.00 m/s to the right as shown. (a) What is the velocity of the wedge after the block reaches the horizontal surface? (b) What is the height h of the wedge?



Q6.

Stretchable ropes are used to safely arrest the fall of rock climbers. Suppose one end of a rope with unstretched length l is anchored to a cliff and a climber of mass m is attached to the other end. When the climber is a height l above the anchor point, he slips and falls under the influence of gravity for a distance of $2l$, after which the rope becomes taut and stretches a distance x as it stops the climber. Assume a stretchy rope which behaves like a spring with spring constant k . Applying the work-energy principle, show that

$$x = \frac{mg}{k} \left[1 + \sqrt{1 + \frac{4k\ell}{mg}} \right].$$



Q7. (Challenging)

The magnitude of the air resistance acting on a cyclist is kv_r^2 , where v_r is the velocity of the air relative to the cyclist and k is a constant. The force acts in the direction of the relative velocity.



The figure shows a cyclist travelling with a road speed v (i.e. speed relative to the road). The wind speed is w and it is at an angle θ relative to the direction in which the cyclist is moving, as shown in the figure.

- (a) What is the power developed by the cyclist in overcoming a (i) head wind ($\theta = 0$); (ii) cross wind ($\theta = 90^\circ$)? (Before you calculate, ask yourself, intuitively or from experience, should the two answers be equal? Or should one answer be greater than the other? If so, which one?)
- (b) Show that the power required to cycle at speed v in a cross wind of speed v is a factor of $\sqrt{2}$ greater than the power required to cycle at the same speed v in still air.

Answers

Q1. 2.5 m s^{-2} , 6.3 m s^{-1}

Q2. 82.3 N , 4.5 m s^{-2}

Q3. 1.7 m s^{-2} , 430 N ; 1.7 m s^{-2} , 220 N

Q4. (a) $a_2 = 2 a_1$, (b) $T_1 = m_1 g - m_1 a_1$, $T_2 = m_2 a_2$ (c) $a_1 = \frac{m_1 g}{m_1 + 4 m_2}$, $a_2 = \frac{2 m_1 g}{m_1 + 4 m_2}$

Q5. -0.667 m/s , 0.952 m

Q7. $kv(v+w)^2$, $kv^2(v^2 + w^2)^{\frac{1}{2}}$